An Explainable Driving Behavior Analytics Framework for Government-Aligned Pay-As-You-Drive Models Using **PiViTel Edge Telemetry**

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1 Background and Policy Context

India is rapid digital transformation in transportation systems under national initiatives such as the **Digital India Mission** [1] and the **Ministry of Road Transport & Highways (MoRTH)** Intelligent Transportation Systems (ITS) programs [4]. These initiatives promote smart mobility, digital governance, and data-driven infrastructure development.

In 2022, the **Insurance Regulatory and Development Authority of India (IRDAI)** enabled telematics-based motor insurance add-ons including *Pay As You Drive (PAYD)* and *Pay How You Drive (PHYD)* models. Official notification: [2].

These models shift traditional fixed premium insurance structures toward behavior-based and usage-based pricing, aligning with:

- Fair and risk-sensitive premium determination
- Road safety incentivization
- Digital transparency in financial decision-making
- Data-driven governance frameworks

2 Road Safety Motivation

According to the 'Road Accidents in India' report published by MoRTH (2022) [3], over 4.6 lakh road accidents were reported in India, with overspeeding contributing to a significant proportion of fatalities.

This highlights the urgent need for behavior-aware monitoring systems that can:

- Detect risky driving patterns: Unsafe driving behaviours like frequent overspeeding, harsh braking and rapid acceleration.

- Quantify exposure to road hazards: Keeps an eye on where and when the driver drives by measuring the mileage or driving during peak traffic and time spent on accident-prone roads.

- Incentivize safer driving practices: Creates a financial motivation for drivers to improve their driving practices by receiving lower premiumns or discounts for the drivers who maintain speed limits, or aggressive driving.

PAYD and PHYD models therefore function not only as insurance innovations but also as behavioral intervention mechanisms supporting national road safety goals.

3 Observational Gap

PiVitel is an edge-based vehicular IoT system designed to collect structured telemetry data including:

- Frame number and timestamp
- Vehicle speed
- Object detection confidence
- Object class
- Distance to detected object
- GPS latitude and longitude
- Spatial object position

While the system efficiently captures high-volume structured CSV data, the following analytical gaps exist:

- Raw telemetry does not directly represent driving quality or risk exposure.

- PAYD frameworks require derived behavioral indicators such as overspeed frequency, sudden acceleration, close-proximity events, and high-risk zone exposure.

- Absence of explainable analytics reduces transparency and stakeholder trust.

Thus, a critical gap exists between IoT data acquisition and policy-aligned driving behavior evaluation.

4 Need for Explainable and Responsible Analytics

Insurance premium determination is a high-stakes financial decision. Therefore, transparency and fairness are essential.

The **NITI Aayog Responsible AI for All** framework [5] emphasizes:

- **Transparency :** The system should be able to explain how and why a particular insurance premium was assigned and see what are the factors influencing the calculation of insurance premium.
- **Accountability:** There must be a clear responsibility for all the decisions made by the systems.
- **Fairness:** Fair premium calculations without be affected by irrelevant or sensitive attributes.
- **Auditability:** Allow independent verification and inspection of the entire process.

Without explainable mechanisms, behavior-based insurance models risk:

- **Algorithmic bias :** Incomplete training data or unbalanced datasets might lead to the model producing systematically unfair outcomes for a group of users.
- **Reduced user trust:** For an innovative system like this, it might take time for the users to gather trust on the system's decisions.
- **Regulatory concerns:** Concerns like system errors, data security, cost and potential misuse of personal data might raise significant concerns in the privacy.

Hence, a structured, interpretable analytics pipeline is necessary for policy-aligned PAYD deployment.

5 Stakeholder and Empathy Analysis

5.1 Drivers

Drivers may experience concerns regarding fairness, privacy, and lack of clarity in score computation. They require transparent dashboards that:

- Explain score derivation
- Highlight risky events
- Provide improvement suggestions

5.2 Insurance Providers

Insurance companies require:

- Reliable behavioral risk metrics
- Fraud detection capability
- Audit-ready scoring mechanisms

5.3 Government and Regulators

Regulatory authorities require:

- Explainable analytics
- Bias mitigation
- Policy-aligned implementation

5.4 Fleet Operators

Fleet managers require comparative analytics, anomaly detection, and safety performance tracking to optimize operations.

6 Problem Statement

The Government of India has enabled behavior-based insurance frameworks such as PAYD and PHYD. These models require accurate, explainable, and transparent assessment of driving behavior.

Although PiVitel generates rich vehicular IoT telemetry in structured CSV format, raw data does not directly provide behavioral risk indicators, safety metrics, or interpretable driving scores required for fair PAYD implementation.

The absence of a structured data analytics and visualization framework limits:

- **Risk quantification** : Raw telemetics are unable to transform into meaningful risk scores which makes it difficult the probability of accidents accurately and compare risk levels across drivers.
- **Behavioral abstraction** : Decision-making becomes difficult as the driving signals (speed, braking, acceleration) in the raw data are not able to indicate the driving patterns.

- **Transparency in premium determination** : Lack of explainability of raw data makes it difficult for the user to understand the factors affecting the premium.
- **Stakeholder trust** : Without proper visualization the drivers, insurers and regulators might question the fairness and reliability of the models.
- **Regulatory auditability** : A proper inspection and verification of system is required which is possible only with well-defined analytics and visualiztion framework.

Therefore, this project proposes the design and implementation of a comprehensive, explainable, and policy-aligned analytics framework that transforms PiVitel telemetry into actionable driving intelligence supporting Government-promoted PAYD models.

7 Research Question

How can raw PiViTel-generated vehicular IoT telemetry be transformed into explainable, risk-sensitive, and policy-aligned driving behavior metrics that support transparent Pay-As-You-Drive insurance models in India?

8 Initial Data Source

The primary data source for this study is the in-house edge device **PiViTel**, developed by **K-Means Technologies Private Limited**. The device generates structured vehicular telemetry data in the form of CSV files during vehicle operation. For the purpose of this project, full access to the PiViTel-generated datasets will be available, allowing comprehensive data processing, analysis, and visualization required to achieve the objectives of the study.

9 Data Analytics Methods

9.1 Dataset Overview and Validation

1. **Dataset Preparation and Integration**

   From PiViTel, the data was available in two separate sources: video-based object detection data and sensor-based telemetry data as csv file. These datasets were merged using the common attribute `frame_number`.

   During this process, column names were standardized, duplicate records were removed, and timestamp formats were unified. The merged dataset was then sorted chronologically to preserve the actual driving sequence.

The final dataset was stored as:

`data/processed/pivitel_merged_dataset.csv`

2. **Dataset Inspection and Validation**

Before proceeding with further analysis, the dataset was carefully inspected to understand its structure, verify data types, and ensure consistency.

Dataset Overview

Dataset File:

`data/processed/pivitel_merged_dataset.csv`

Metric	Value
Total Rows	4398
Total Columns	47

The dataset contains telemetry observations generated by the PiViTeL edge device, including object detection results, GPS coordinates, IMU sensor data, and temporal metadata.

Dataset Attributes

The dataset consists of the following groups of attributes:

Detection and Tracking Attributes

- frame_number, detection_id, tracking_id
- class, confidence
- bbox_x, bbox_y, bbox_w, bbox_h
- position, distance_m

Timestamp and System Metadata

- manual_timestamp, wall_time_iso
- overlay_time, overlay_date
- timestamp, system_time
- video_time_seconds

GPS Telemetry

- latitude, longitude, gps_altitude
- gps_speed_kmh, gps_course_degrees
- gps_fix_quality, gps_satellites
- gps_hdop, gps_data_age_seconds

IMU Sensor Data

- imu_accel_x, imu_accel_y, imu_accel_z
- imu_accel_magnitude
- imu_gyro_x, imu_gyro_y, imu_gyro_z
- imu_gyro_magnitude
- imu_mag_x, imu_mag_y, imu_mag_z
- imu_mag_magnitude
- imu_temperature, imu_data_age_seconds

Overlay Metadata

- overlay_time, overlay_date, overlay_gps
- overlay_accel_g, overlay_gyro_deg_s
- overlay_ok, overlay_conf, overlay_ms

Data Type Verification

The dataset contains multiple data types:

Data Type	Example Columns
Integer	frame_number, gps_fix_quality, gps_satellites
Float	confidence, GPS values, IMU readings
Boolean	overlay_ok
String	class, position, timestamps

This combination of data types is consistent with real-world telemetry systems.

Missing Value Analysis

Missing values were analyzed across all attributes.

Key Observations:

(a) Object detection related attributes contain missing values in **1098 rows**, including detection ID, tracking ID, class, confidence, bounding box coordinates, position, and distance. These correspond to frames where no object was detected.

(b) Several overlay-related attributes contain **100% missing values (4398 rows)**:

- overlay_time, overlay_date
- overlay_gps, overlay_accel_g
- overlay_gyro_deg_s

(c) GPS attributes contain minor missing values:

Column	Missing Values
latitude	69
longitude	69
gps_altitude	69
gps_hdop	69
gps_data_age_seconds	69

(d) GPS speed and course values contain moderate missing values:

Column	Missing Values
gps_speed_kmh	485
gps_course_degrees	485

(e) IMU sensor data contains no missing values, indicating reliable recording.

Statistical Summary

Basic statistical analysis confirmed that the data falls within expected ranges:

- Frame numbers range from 1 to 2616
- Detection IDs range from 1 to 8
- Tracking IDs range from 1 to 121
- Temperature values range approximately from 18°C to 45°C

These values are consistent with real-world vehicle telemetry.

Conclusion of Validation

The dataset structure, data types, and attribute distributions were verified successfully. Although missing values are present, they are expected due to sensor behavior and object detection conditions.

Overall, the dataset is suitable for further analysis, including cleaning, feature engineering, and behavioral modeling.

3. **Missing Value Treatment**

   Based on the validation analysis, missing values were handled carefully:

   - Completely empty overlay columns were removed
   - Object detection gaps were encoded as meaningful values (e.g., "no_object")
   - GPS values were filled using interpolation
   - Speed and direction were filled using forward and backward fill
   - Timestamps were standardized into datetime format

   The cleaned dataset was saved as:

   `data/processed/pivitel_cleaned_dataset.csv`

4. **Trip Timeline Reconstruction**

   The dataset was sorted using timestamps to reconstruct the driving sequence. A new feature `time_gap` was calculated to detect irregularities in data capture and ensure temporal consistency.

5. **Feature Engineering**

   To convert raw telemetry data into meaningful behavioural indicators, several features were created:

   - Time difference between frames
   - Speed and acceleration changes
   - Overspeed detection
   - Sudden braking and acceleration events
   - High acceleration detection
   - Object presence indicator
   - Close object risk
   - Trip distance approximation
   - Time-based features (hour, night driving)

   These features enable detailed analysis of driving behaviour and risk patterns.

   The final dataset was stored as:

   `data/processed/pivitel_feature_dataset.csv`

9.2 Exploratory Data Analysis (EDA)

9.2.1 Descriptive Analysis

To understand the overall driving behaviour captured in the dataset, descriptive statistics were computed for key variables such as speed, acceleration, and object distance. This helped in identifying general driving patterns, variability, and potential risk indicators.

Central Tendency:

The average vehicle speed was observed to be approximately **33.03 km/h**, while the median speed was slightly higher at **34.97 km/h**. This indicates that most of the driving occurred at moderate speeds, with no significant skew towards extreme values.

The average acceleration recorded was **9.92**, which remains fairly stable. This is expected, as IMU acceleration values typically include the effect of gravity, resulting in a narrow range of variation.

The average distance to detected objects was found to be approximately **29.61 meters**, suggesting that, on average, the vehicle maintained a reasonable distance from surrounding objects.

Variability Analysis:

The standard deviation of speed was **9.31**, indicating moderate variation in driving speed. This suggests that while the driver mostly maintains a consistent speed, there are occasional changes due to traffic or road conditions.

In contrast, the acceleration standard deviation is very low (**0.21**), which confirms that acceleration values remain stable throughout the trip.

The distance to objects shows a higher variability (**25.41**), indicating that the vehicle encounters both close and distant objects during the journey. This reflects changing driving environments such as traffic density.

Extreme Values:

The maximum speed recorded in the dataset is **53.62 km/h**, while the minimum speed is **2.30 km/h**. This suggests that the vehicle operates within a relatively safe speed range and does not exhibit extreme high-speed behaviour.

Driving Event Analysis:

To understand driving behaviour more clearly, event-based features were analyzed:

- Overspeed events: **93**
- Sudden acceleration events: **2**
- Sudden braking events: **3**
- Close object risk events: **1098**

The corresponding percentages are:

- Overspeed: **2.11%**
- Sudden acceleration: **0.04%**
- Sudden braking: **0.06%**
- Close object risk: **24.97%**

Interpretation:

The results indicate that the driver generally maintains controlled and stable driving behaviour. The relatively low percentage of overspeed events suggests that speed limits are mostly respected.

Sudden acceleration and braking events are extremely rare, which implies smooth driving without aggressive maneuvers.

However, a significant observation is the high percentage of **close object risk events (approximately 25%)**. This indicates that although the driver maintains moderate speed, the vehicle frequently operates in close proximity to surrounding objects. This may be due to dense traffic conditions or urban driving environments and represents a potential safety concern.

Overall, the descriptive analysis suggests that while the driving behaviour is largely stable and controlled, proximity to objects is the primary risk factor that needs further investigation in subsequent analysis stages.

9.2.2 Univariate Analysis

Univariate analysis was carried out to understand the behaviour of individual variables in the dataset without considering their relationships with other features. This involved examining statistical measures such as mean, median, and standard deviation, along with observing the overall distribution of values. Through this step, it became easier to identify typical value ranges, detect any unusual patterns, and understand how different attributes such as speed, acceleration, and object distance behave during the driving session. This provided a clear baseline understanding of the dataset before moving on to more complex analysis.

9.2.3 Bivariate Analysis

Bivariate analysis was performed to explore how different variables interact with each other and to identify relationships that may influence driving behaviour. By comparing pairs of variables such as speed with object distance or speed with risk events, it was possible to observe patterns that are not visible in individual analysis. This step helped in understanding how certain conditions, like higher speeds or closer object distances, relate to risky driving events. Overall, it provided deeper insights into how different factors combine to affect driving safety and behaviour.

9.2.4 Correlation Analysis:

Correlation analysis will be performed to determine the strength of the relationships between numerical variables in the dataset. A correlation matrix will be generated to measure the degree of association between attributes vehicle speed, object distance, and detection confidence. This analysis will help in identifying the variables that may significantly influence the driving behaviours of the people and will help in applying the correct feature engineering methods.

9.3 Behavioral Analytics

1. Feature Engineering: Feature engineering will be applied to transform raw telemetry variables into meaningful behavioural indicators. Feature engineering becomes important for derived features as they are the ones which are going to provide an interpretable representation of the driving behaviour which will help in behavioural risk analysis. The important features include overspeed events, speed variance, proximity risk indicators, and object encounter frequency.

2. Risk Metric Computation: Once the behavioural indicators are computed, risk metrics will be computed based on that that will quantify driving behaviour. These metrics includes overspeed rate, proximity risk score, and speed stability indicators. This evaluation metrics will provide us the level of risk associated with a driver's behaviour based on these telemetry observations.

3. **Driving Score Model** The computed behavioral metrics will give a driving score which represents the safety level of each driver's behaviour. This score will combine behavioural factors such as speed compliance, safe distance maintenance and driving stability to result in a single measure which will further help to study the comparison between different driving ways and provide a proper behaviour-based risk evaluation.

   Time-Series Analysis The telemetry dataset mainly contains timestamped observations which will help to perform a time-series analysis to analyze how the driving behaviour changes during a particular driving session. These changes will further allow us to detect the sudden fluctuations or unstable driving patterns, if any.

   Anomaly Detection Anomaly detection will be used to identify any unusual driving behaviour observed in the dataset. Anomalies might include extremely high vehicle speeds, sudden proximity to objects or irregular sensor readings which on detecting will help identify the potential risks in the driving behaviour.

Spatial Analysis Spatial analysis will be performed in the geographic coordinates captured by GPS which is contained in the telemetry dataset, to analyze the driving behaviours across different locations. The mapping techniques will be helpful to visualize different vehicle routes and identify the locations where there are chances of risky driving behaviour.

10 Visualization Methods Used

To better understand the driving behaviour present in the dataset, a combination of visualization techniques was used at different stages of the analysis. Initially, univariate visualizations were applied to study individual variables in isolation. Histograms were used to observe the distribution of numerical features such as vehicle speed, acceleration, distance to objects, and confidence scores. These plots helped in identifying common value ranges and detecting any unusual patterns. Boxplots were also used, particularly for speed, to identify variability and potential outliers. For categorical and event-based variables such as overspeed, sudden braking, sudden acceleration, and close object risk, bar charts were used to understand their frequency, while pie charts helped in comparing their relative proportions.

In addition to this, time-based and spatial visualizations were included to capture more context. Hour-wise distribution plots were used to analyze when most driving activity occurs, and night driving indicators were used to understand risk during different time periods. GPS-based scatter plots were used to visualize the vehicle's route, with speed represented through color variation to show how driving conditions change across locations. A circular (rose) plot was also used to analyze GPS course direction, providing insight into movement patterns and directional changes during the trip.

After understanding individual features, bivariate visualizations were used to explore relationships between variables. Scatter plots and boxplots were used to study interactions such as speed vs distance and speed vs risk events, helping to identify how different factors contribute to unsafe conditions. More advanced visualizations such as density plots, risk probability plots, correlation heatmaps, and pairplots were used to gain deeper insights into the data. These methods made it easier to identify patterns, relationships, and potential risk factors, which are important for further behavioural analysis and model development.

10.1 Univariate Analysis Visualization

Univariate analysis was performed to examine the distribution and behaviour of individual variables using histograms, boxplots, and bar charts, helping to understand data patterns and detect outliers.

- **Acceleration Distribution:** considers the IMU acceleration magnitude and divides values into ranges to see how frequently each range occurs.

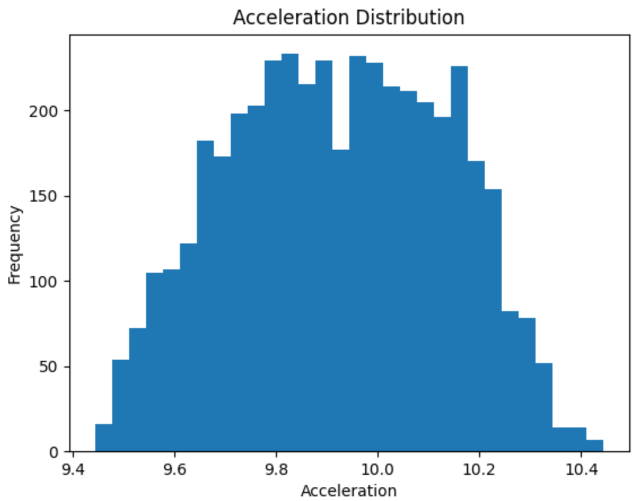

Figure 1: acceleration distribution

The graph shows that the distribution is roughly normal and peak is around $9.8 - 10\,\mathrm{m/s^2}$ depicting that values are close to gravitational acceleration and acceleration is stable.

- **Object Class Pie Chart** counts frequency of each detected object class and displays proportion as percentages.

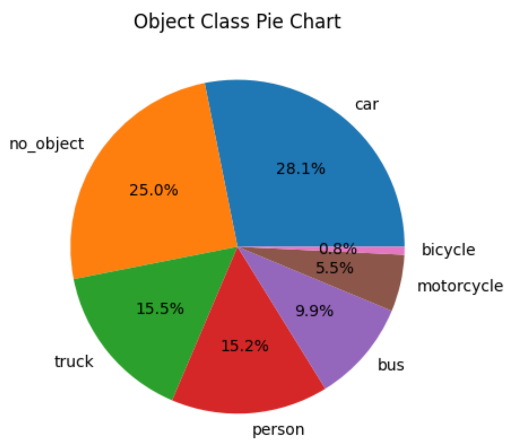

Figure 2: object class pie

Most detections are observed to be vehicles and there are many observations with no objects that shows the presence of many frames without detections.

- **Close Object Risk:** counts the occurrences of no risk or close object risk.

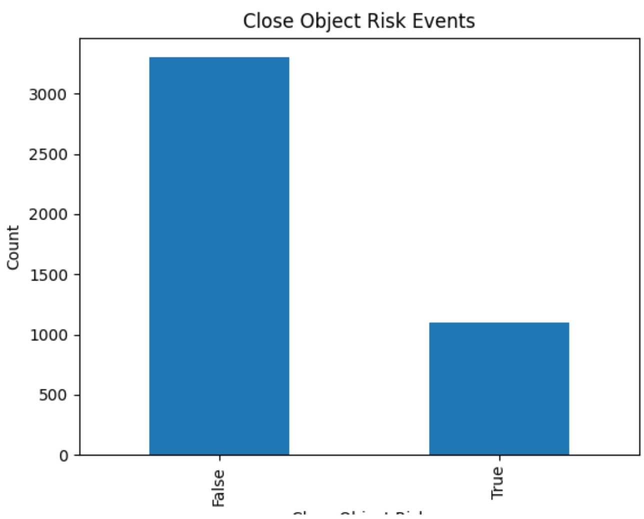

Figure 3: close object risk

Shows a large number of False depicting that mostly driving is safe while a proportion of it involves close proximity which possibly shows the possibility of dense traffic.

- **Model Confidence Analysis:** shows the reliability of the object detection.

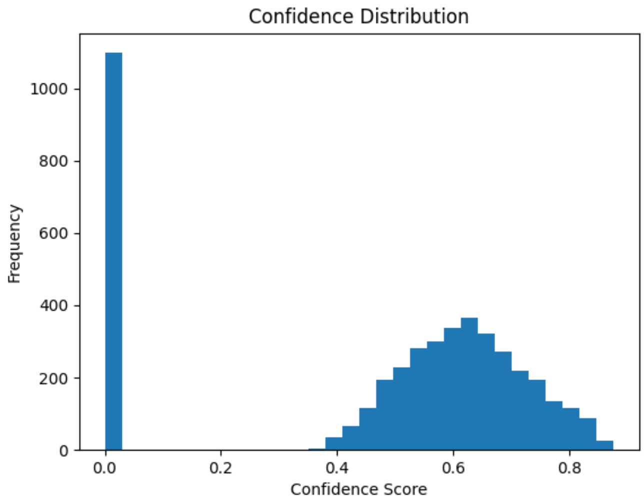

Figure 4: confidence histogram

A significant number of entries have a confidence score of 0.0 which shows frames where no objects were detected. The second cluster peaks around 0.6-0.65 that depicts moderately confidence.

- **Directional Analysis:** analyzes the driving direction patterns and the concentration of movement in the directions.

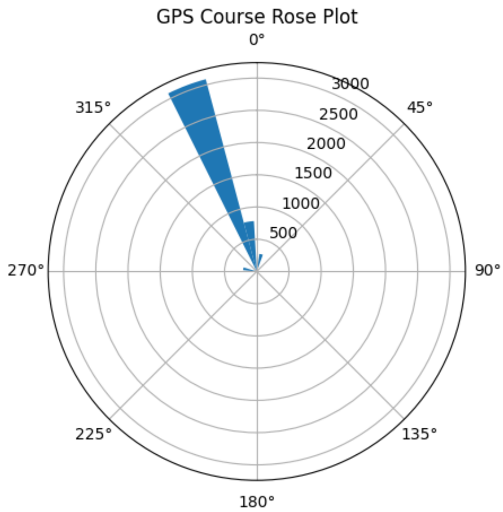

Figure 5: gps course plot

This circular histogram uses the gps course degrees and considers 0 degrees as the North and the angle increases clockwise. The large "petal" pointing towards 330 degrees (North-Northwest) indicates that the vehicle spent the vast majority of this dataset traveling in that specific direction.

-**Spatial and Velocity Mapping**: visualizes the speed variations along the route and where the vehicles are getting moved based on the vehicle path based on the GPS route.

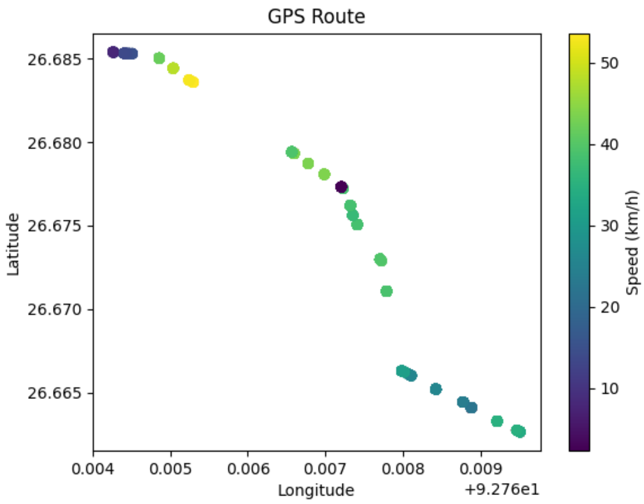

Figure 6: acceleration distribution

The color mapping shows dark purple dots indicating low speeds (0-10 km/h), yellow dots indicate speeds over 50 km/h. The plot shows that the vehicle accelerated at the start of the route and maintained a steady speed and slows down towards the bottom right.

- **Safety Event distribution:** depicts a plot based on high acceleration distribution.

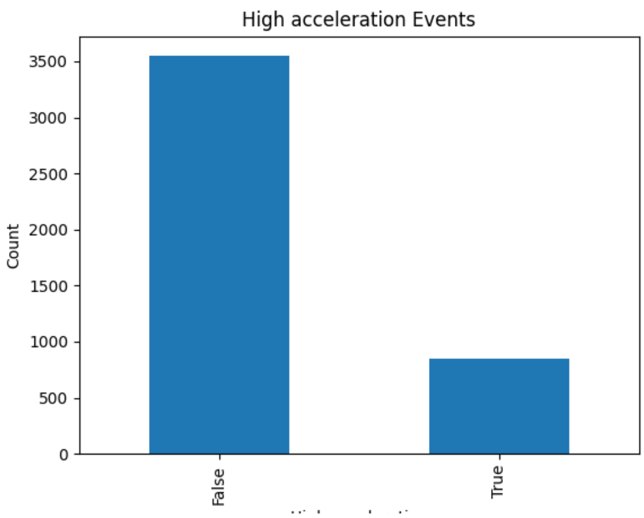

Figure 7: high acceleration distribution

The graph shows 3500 instances of normal driving and about 800 instances of high acceleration showing different driving styles of the drivers.

- **Time-based analysis (hour of day):** shows the time when driving occurs mostly.

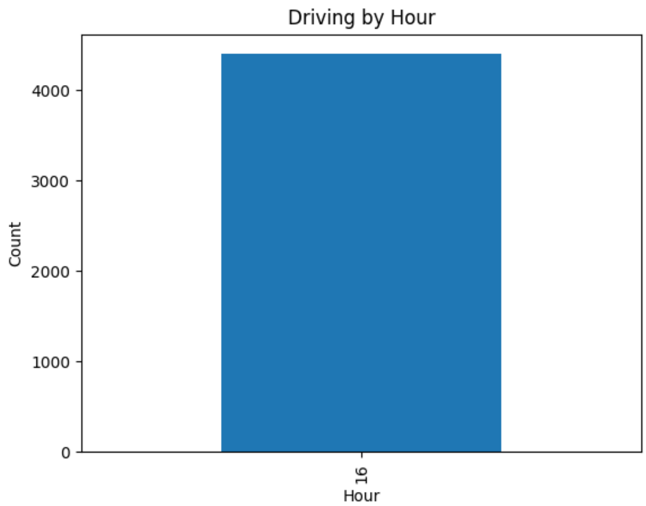

Figure 8: hour of day

The plot shows that the collection of datapoints are recorded or shows mostly data have been collected during 16:00 hour.

- **Time-based analysis (night time):** counts rows with day driving (denoted by 0) or night driving (denoted by 1)

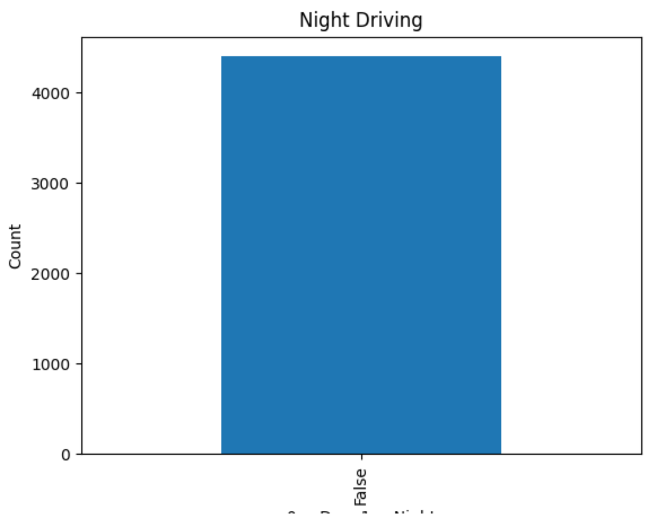

Figure 9: night driving

The graph shows only one bar with no True(1) values indicating that the dataset contains only daytime driving data.

- **Risk events by hour:** groups data by hour_of_day and sums all risk events per hour in a bar chart.

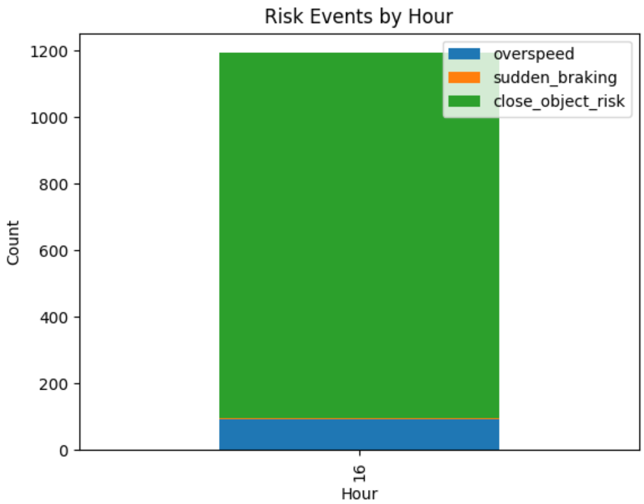

Figure 10: risk events by hour

The dataset was recorded only at 16:00 hour so there are no variation accross time and all risk behaviours are concentrated in a single time window. The blue color depicts overspeed (small portion), orange for sudden braking (very small) and green for close object risk (dominant).

- **Risk Event distribution:** shows the distribution of risk events in piechart.

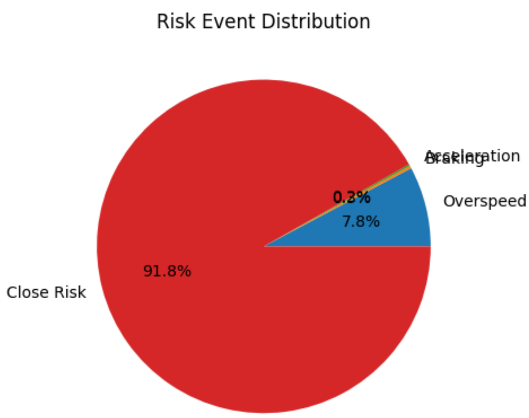

Figure 11: risk events distribution

The pie chart shows that close risk covers majority portion with 91.8% followed by over-speed and a very small portion for braking.

- **Vehicle Speed Boxplot:** The boxplot shows median speed, spread of data and outliers.

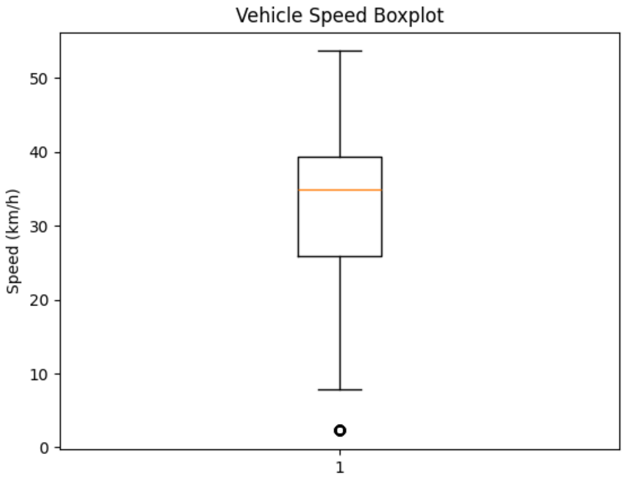

Figure 12: vehicle speed boxplot

The median speed is observed to be 35 km/h and most speeds are between 25 to 40 km/h. There are very few extremely high speeds and there is atleast one very low speed recording (which might be possible when the vehicle was stopped or moving very slowly)

- **Vehicle speed distribution**: shows the distribution of speed.

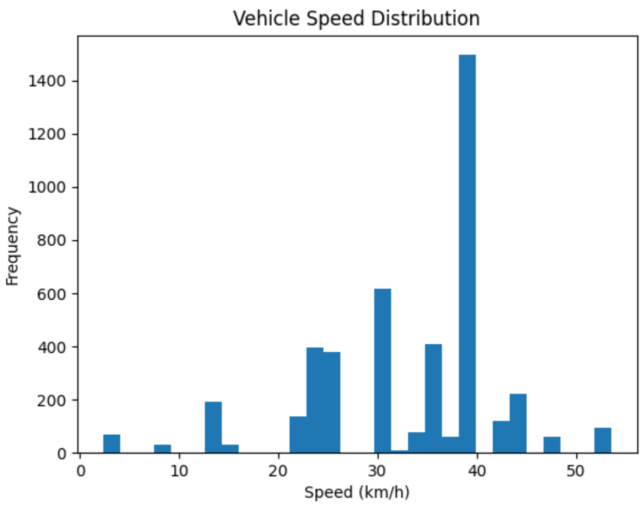

Figure 13: speed distribution

The distribution is heavily concentrated around 35-40 km/h with the tallest bar is at 37-38 km/h and small peaks at lower speeds (around 20 km/h) possibly suggesting slow-moving areas or traffic stops.

- Sudden Acceleration events:

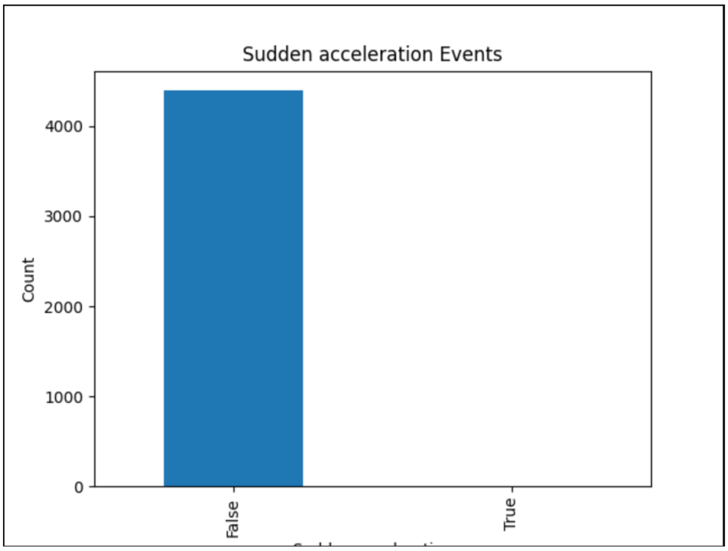

Figure 14: sudden acceleration events

There are many data points where sudden acceleration did not happen which depicts the driver is extremely smooth while accelerating.

- Sudden Braking events:

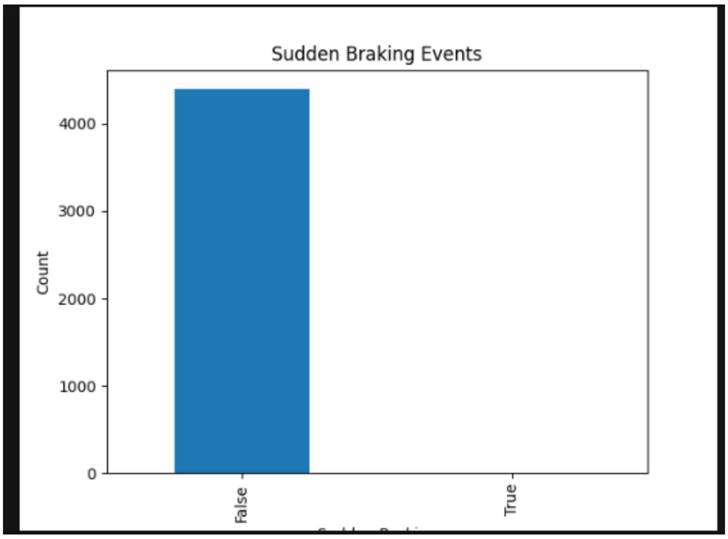

Figure 15: sudden braking events

The plot shows braking events are rare depicting that the driver is generally smooth while braking.

- **Trip distance:** shows travel progression and distance variation

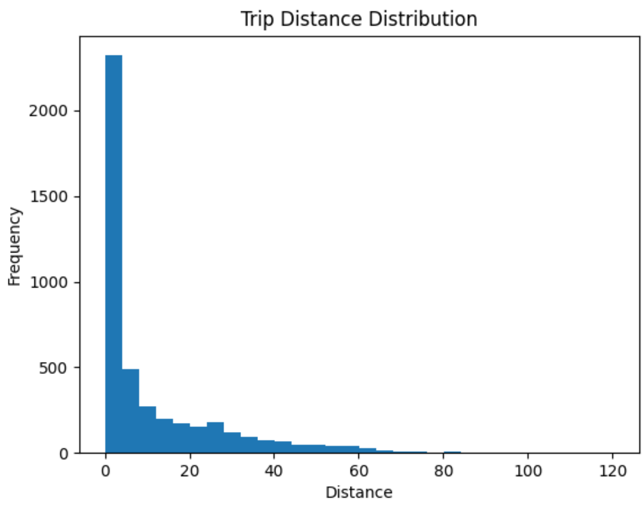

Figure 16: trip distance

The graph shows majority of trips are very short which means the data is mostly from daily commuting and very few trips longer than 40 km.

10.2 Bivariate Analysis Visualization

Bivariate analysis was used to study relationships between variables through scatter plots, boxplots, and advanced visualizations, enabling identification of interactions and potential risk patterns in driving behaviour.

- **Acceleration vs Close Object Risk**:

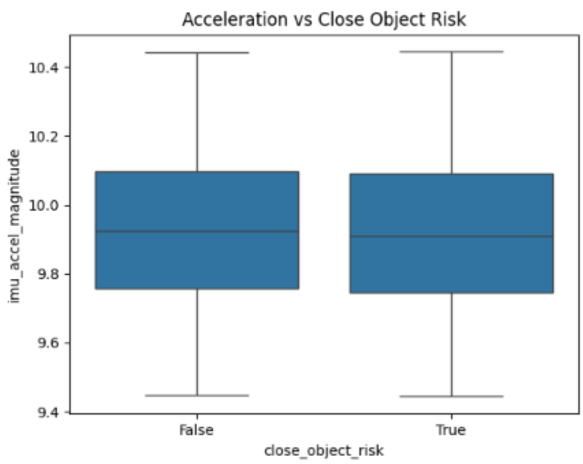

Figure 17: Acceleration vs Close Object Risk

The boxplot shows both boxes with almost identical shape, median and spread indicating no significant difference in acceleration magnitude.The median acceleration is around 9.95 in both the plots.

- **Distance vs Sudden braking**:

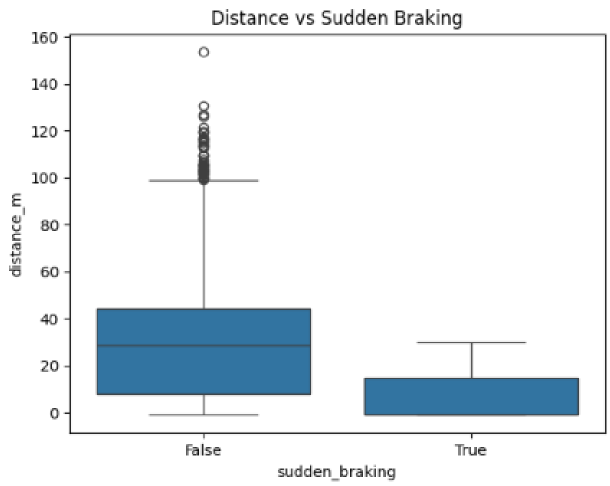

Figure 18: Distance vs Sudden braking

The x-axis shows sudden braking (false or true) and y-axis shows distance to object. The False (no sudden braking) shows median distance (30-35 meters) with wide spread (0-100 m) and several high outliers up to 150 m. The True (sudden braking occurred) shows much lower median distance (10 - 15 meters) with much narrower box and no extreme outliers.

- **Acceleration vs Braking**:

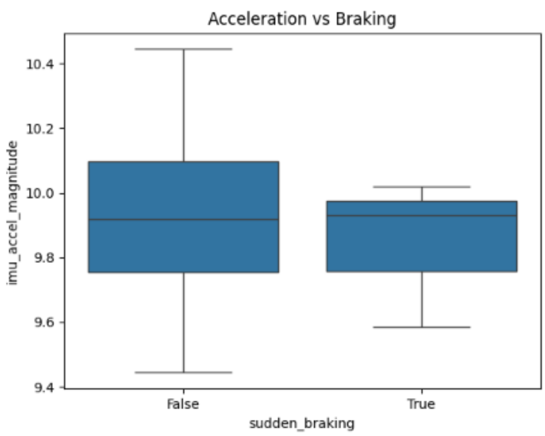

Figure 19: Acceleration vs Braking

The boxplot shows False with slightly higher median acceleration and True with lower median acceleration but there is almost no relationship between current acceleration and whether sudden braking occurs.

- **Close Object Risk by hour**:

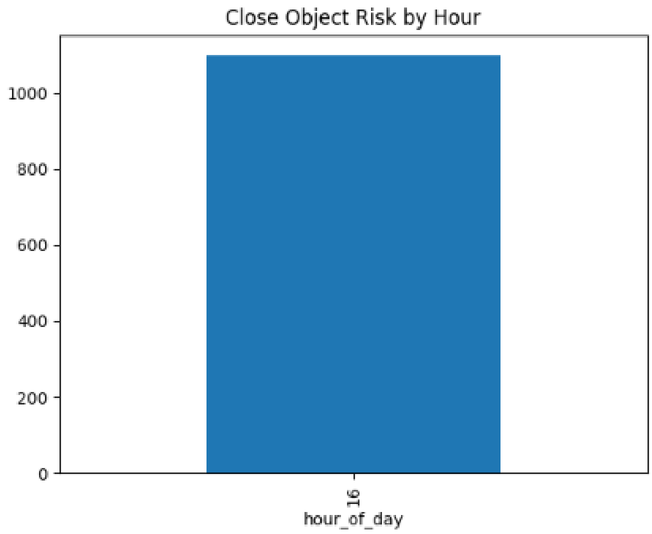

Figure 20: Close object Risk by hour

The graph depicts a single bar at 16 hour with a count of approximately 1100 depicts close objects are highly concentrated at this hour.

- **Risk Event Correlation Heatmap**:

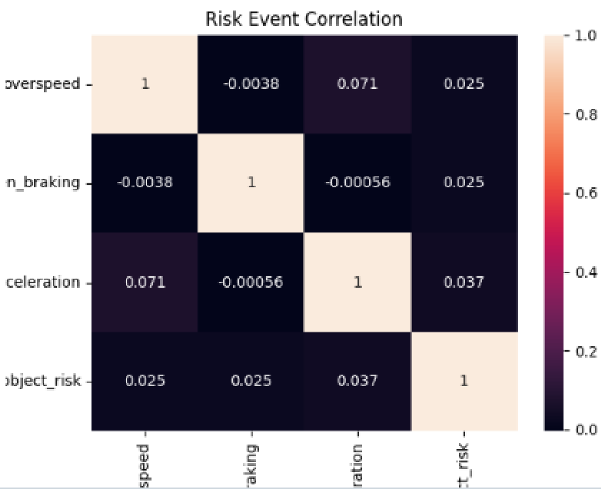

Figure 21: Risk Event Correlation Heatmap

The heatmap shows that correlations are very low (mostly between -0.005 to 0.071) indicating no strong relationships between any risk events (overspeed, sudden braking, sudden acceleration, close).

- **Speed vs Distance Density:**

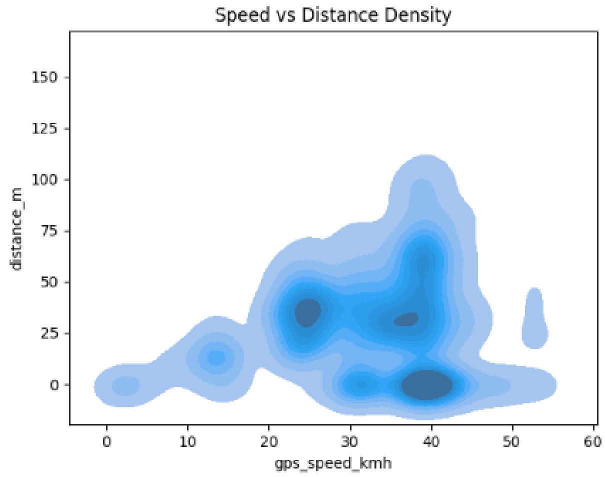

Figure 22: Speed vs Distance Density

The plot shows dark blue regions that have higher density of data points. The highest density occurs mainly around 30-40 km/h and distance of 0-50 meters (where objects are relatively close) indicating probably urban style driving.

- **Distance vs Object Detection**:

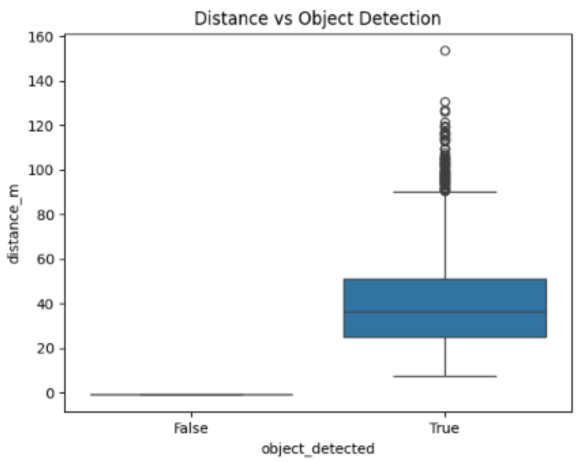

Figure 23: Distance vs Object Detection

The x-axis denotes object detected (false vs true). The False shows distance is almost zero and the True shows much higher distances of median 35-40 meters with points upto 90-130 meters indicating that distance is meaningful only when an object is detected.

- **Pairplot**:

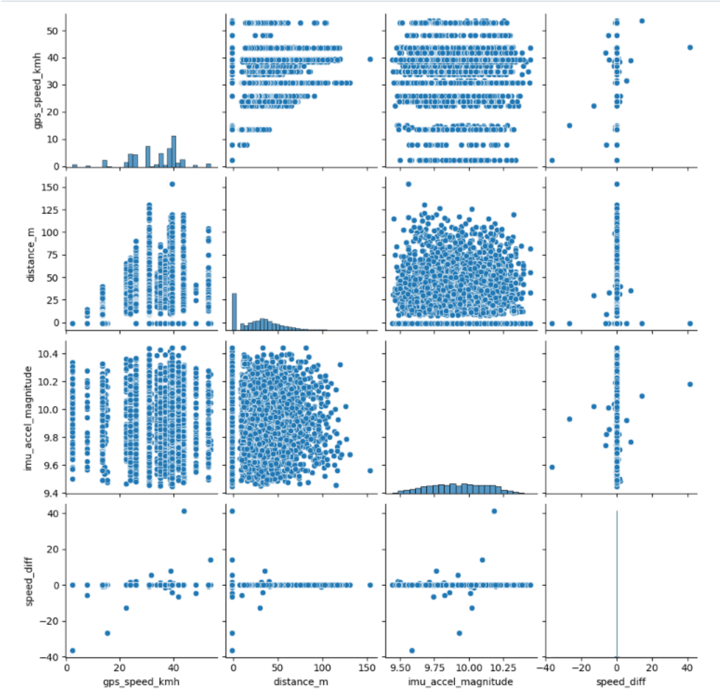

Figure 24: Pairplot

The plots shows there is a strong concentration of data in moderate speeds. The acceleration values are consistent while the distance and speed shows variation in the dataset.

- **Risk probability vs Speed**:

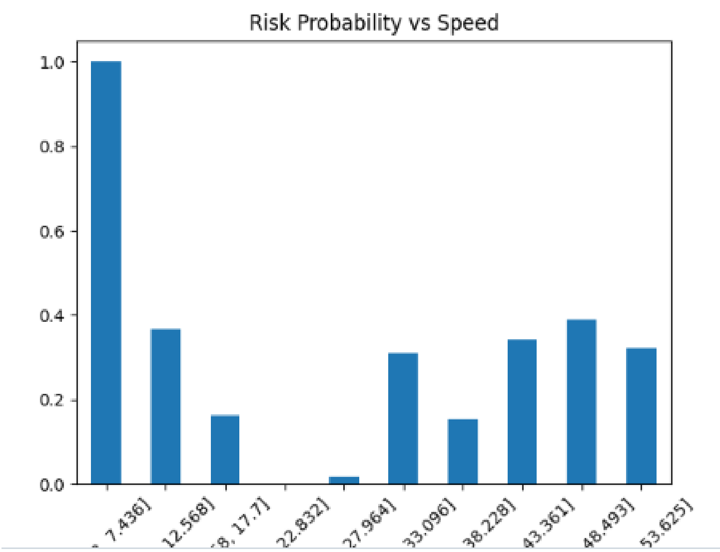

Figure 25: Risk probability vs Speed

The graphs shows extremely high risk at very low speeds (0-7 km/h) as close object risk is almost present always. The risk drops in the 18-22 km/h range and rises again at around 35-40 km/h. This indicates non uniform distribution of close object risk.

- **Speed vs Sudden Acceleration**:

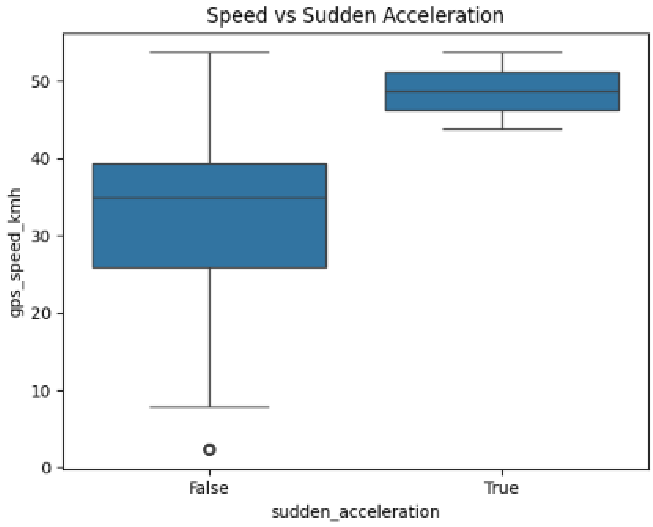

Figure 26: Speed vs Sudden Acceleration

Both the boxplots looks similar. The x-axis shows whether sudden acceleration is present or not. No sudden acceleration is observed around speed range of 35km/h while when speeds occurs consistently higher with median around 48-50 km/h, sudden acceleration events happens (suggesting that driver accelerates harshly when they are already moving at a higher speed).

- **Speed vs Acceleration**:

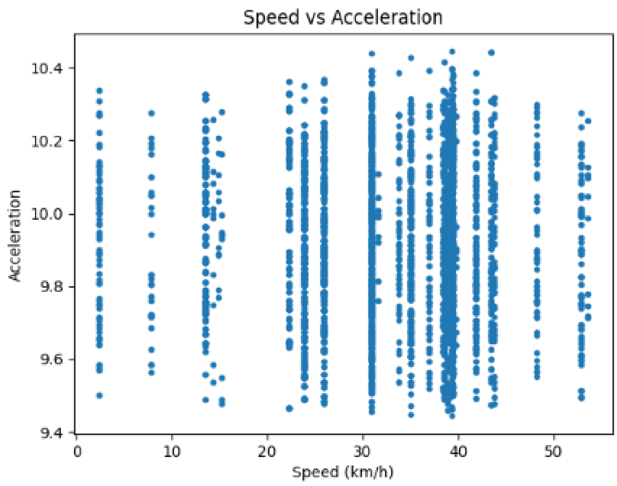

Figure 27: Speed vs Acceleration

Acceleration values are clustered tightly around 9.5 and 10.4 and no clear linear trend is visible as points are scattered across all speeds. A slight variation in acceleration is seen at higher speeds and at very low speeds, acceleration is seen to be more variable.

- Speed vs Sudden Braking:

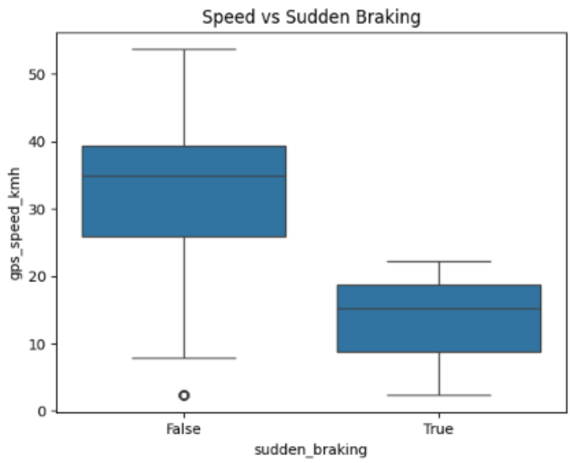

Figure 28: Speed vs Sudden Braking

The boxplot shows two conditions- no sudden braking occurs and when sudden braking occurs. No sudden braking is observed in median speed around 35 km/h while sudden braking occurs when median speed drops to around 16-18 km/h (lower speeds) indicating that drivers tend to brake suddenly in low speed conditions like traffic or when an object suddenly appears at close range.

- Speed vs Distance:

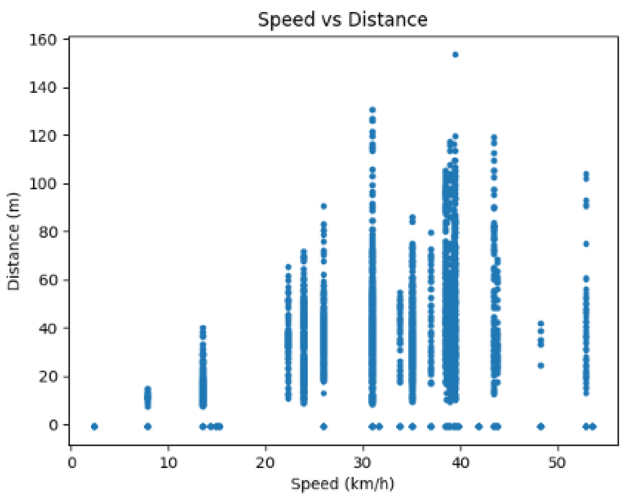

Figure 29: Speed vs Distance

The scatter plot shows low speeds (¡15 km/h) at low distance (0-20m), moderate speeds (25-40 km/h) at various distance ranging from 0 to 130 meters, and higher density of points are around 35-42 km/h with distances between 20-100m. Very few points are observed at speed above 35 km/h.

- **Speed vs Close object risk**:

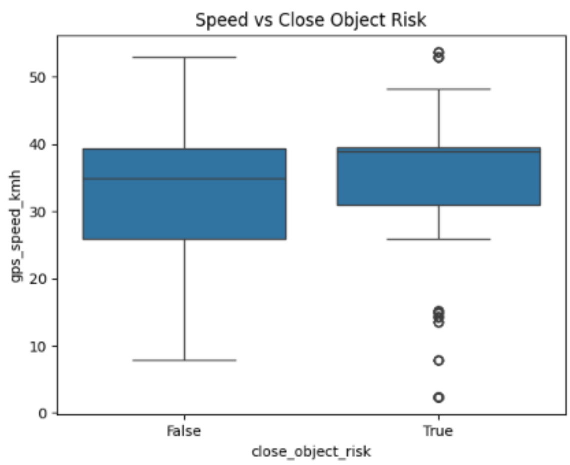

Figure 30: Speed vs Close Object Risk

The plot shows no close risk at median speed of 35 km/h. At median speed of 37-38 km/h close object risk is present.

- Speed vs Object Class:

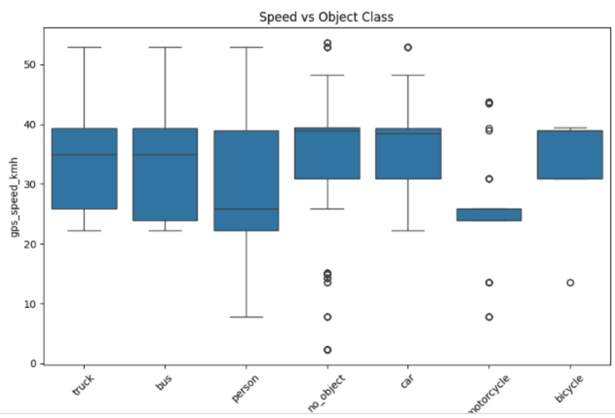

Figure 31: Speed vs Object Class

The plot explains that the driver appears to slow down (narrow box) the speed to 25 km/h on detecting motorcycles and person while indicating truck, bus, car and bicycles shows an average speed of around 35-38 km/h. In case of person, a wider variation is observed.

- **Speed vs Hour of day**:

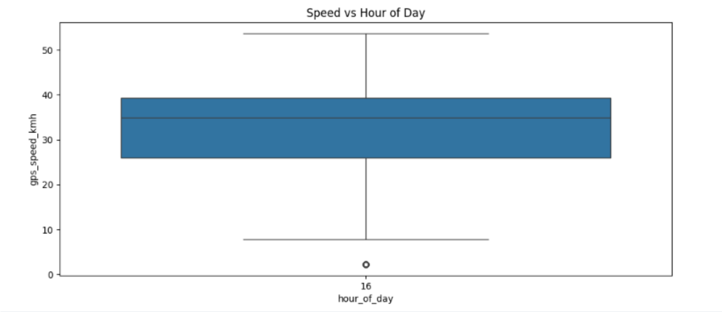

Figure 32: Speed vs Hour of day

The plot indicates that the dataset contains driving data primarily from 16:00 hour with a median speed of approximately 35-36 km/h and a single low outlier is observed near 2-3 km/h (which might indicate the vehicle stopped at traffic light, in congestion or might have paused.)

10.3 Correlation Visualization

A correlation heatmap will be generated to measure the strength of relationships between numerical variables in the dataset that will visually represent correlation coefficients between variables such as vehicle speed, object distance, and detection confidence.

This visualization will help in identifying variables that may influence driving behaviour and help applying proper feature engineering techniques for behavioural risk analysis.

10.4 Behavioural Risk Visualization

To visualize the behavioural risk, bar charts is being used to display the frequency of overspeed events and occurences of proximity risks based on the behavioural indicators such as overspeed events, proximity risk indicators, speed variance. Histograms will also be generated to visualize the distribution of computed driving scores that will help categorize driving behaviour into different safety levels and provide a clearer interpretation of driver risk profiles.

10.5 Temporal Driving Behaviour Visualization

Line plots is used to analyze changes in vehicle speed over time. These plots will help observe how driving behaviour evolves during a driving session and identify parts where aggressive driving behaviour may occur.

10.6 Spatial Visualization of Driving Routes

The GPS coordinates allows the spatial visualization helping to analyze geographical driving patterns using mapping libraries such as Folium, vehicle routes can be plotted based on the latitude and longitude data.

Additionally, locations where risky driving behaviour occurs (such as overspeed events) will be highlighted on the map that will allow identification of geographical areas where driving risk may be higher.

10.7 Explainable Driving Score Visualization

Finally, visualizations will be used to interpret the driving score model developed in the project. Feature importance plots will be generated to illustrate how different behavioural indicators contribute to the driving score.

These visualizations will support explainable analysis by clearly demonstrating which driving factors most significantly influence driver safety evaluation.

Overall, multiple visualization techniques including histograms, box plots, scatter plots, pair plots, correlation heatmaps, line charts, bar charts, and spatial route maps will be used in this project. The use of diverse visualization methods will enable comprehensive exploration of the telemetry dataset and facilitate effective communication of analytical insights related to driving behaviour and risk evaluation.

11 Team Contributions and Collaboration

This project was carried out collaboratively with a structured approach to task division, version control, and continuous integration. Each team member contributed to different stages of the analytics pipeline while maintaining coordination using GitHub for proper tracking and management.

11.1 Contribution of Team Members

Jenish A. Borah

Jenish contributed to the core analytical and data processing stages of the project. The contributions include:

- Collection and organization of raw telemetry data generated from the PiViTeL edge device.

- Dataset validation by inspecting structure, verifying data types, and analyzing missing values. A detailed validation report was prepared.
- Implementation of missing value treatment strategies including removal of irrelevant columns, handling detection gaps, interpolation of GPS data, and maintaining temporal consistency.
- Performing descriptive analysis to understand driving behaviour using statistical measures.
- Conducting bivariate analysis to study relationships between variables and identify risk patterns using advanced visualizations.

Shruti Sarma

Shruti focused on data preparation and feature engineering stages of the project. The contributions include:

- Merging multiple raw CSV files into a single unified dataset.
- Performing missing value investigation and categorization to understand data gaps.
- Implementing feature engineering techniques to derive behavioural indicators such as overspeed, sudden braking, acceleration, and close object risk.
- Conducting univariate analysis using various visualizations to understand distribution of individual features.

11.2 Contribution Summary Table

Project Stage	**Jenish A. Borah**	**Shruti**
Data Collection	Collected telemetry data	–
Dataset Merging	–	Merged multiple CSV files
Dataset Validation	Structure validation and report	–
Missing Value Investigation	–	Categorized missing values
Missing Value Treatment	Data cleaning and interpolation	–
Feature Engineering	–	Created behavioural features
Descriptive Analysis	Statistical analysis	–
Univariate Analysis	–	Distribution visualizations
Bivariate Analysis	Relationship analysis	–

11.3 Project Management using GitHub

The project was managed using GitHub to ensure proper version control, collaboration, and task tracking. The repository used is:

`https://github.com/K-Means-Technologoies-Pvt- Ltd/Behavior_Analytics_Framework_of_PiViTeL`

Each stage of the project was organized using GitHub issues, where tasks such as dataset merging, missing value analysis, feature engineering, and EDA were clearly defined.

- Separate branches were used for individual tasks to allow parallel development.
- Pull requests (PRs) were created for each completed task and reviewed before merging.
- Issues were opened and closed to track progress and completion of tasks.
- Commit history was maintained to ensure transparency and traceability.

This workflow ensured proper coordination between team members and maintained a clean and organized codebase.

11.4 GitHub Workflow Evidence

GitHub activities such as issue tracking, pull requests, and commit history were used as supporting evidence of team contributions and project progress. This includes:

- Creation and closure of issues for each project stage
- Pull request submissions and merges
- Branch-based development workflow
- Continuous updates through commits

These practices ensured that the development process was structured, reproducible, and aligned with real-world software development standards.

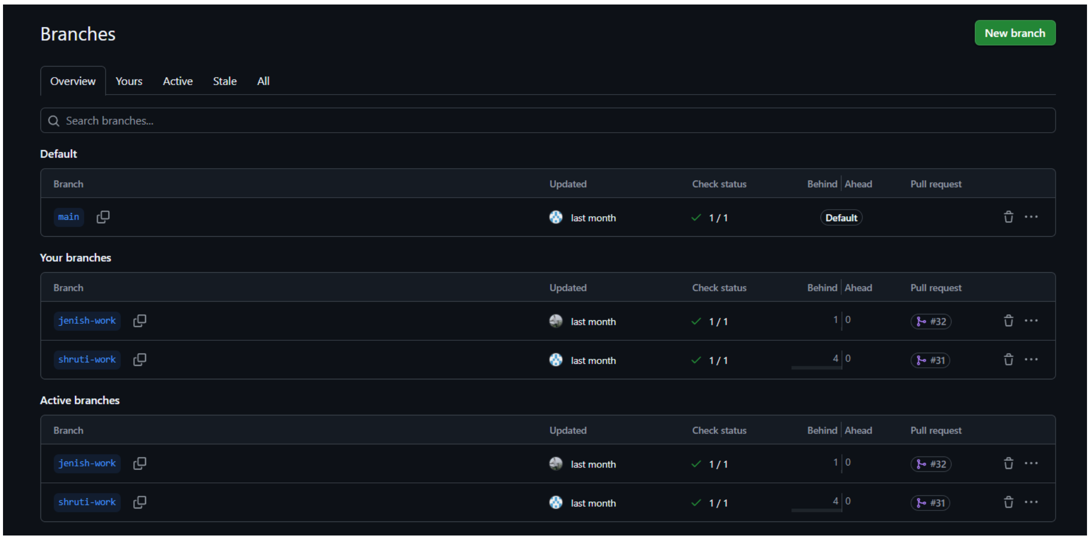

Figure 33: GitHub repository branches showing collabration structure

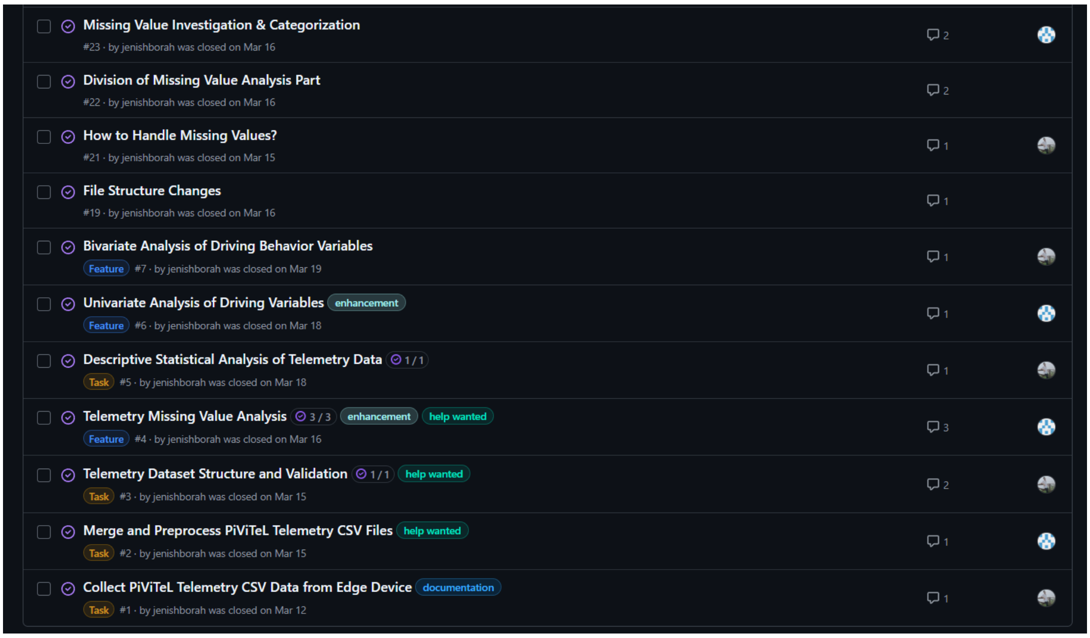

Figure 34: GitHub issue tracking used for task management and progress tracking

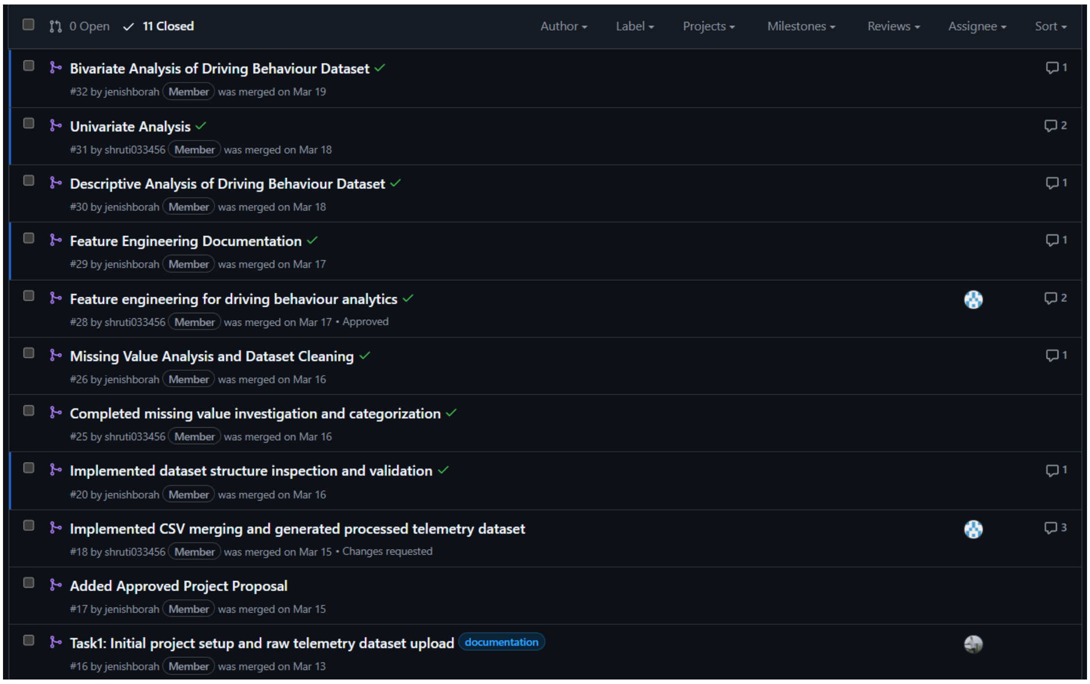

Figure 35: Pull request workflow used for integrating contributions into the main branch

References

[1] Digital India. Digital india mission, 2026.

[2] Insurance Regulatory and Development Authority of India. Document detail (document id: 394286), 2026.

[3] Ministry of Road Transport and Highways. Road accidents in india, 2022.

[4] Ministry of Road Transport and Highways. Intelligent transportation systems (its) programs, 2026.

[5] NITI Aayog. Responsible ai for all, 2026.